Smart FYP – Final Year Project Management System (FYPMS)

Functional Requirements (Organized by Module)

Note: SAP ID is treated as the primary unique identifier for users (students, teachers, committee members, and administrators) and is used across authentication, membership validation, submissions, and reporting workflows.

MODULE 1: Authentication & User Account Management

- FR1.1.1: The system shall allow students to log in using username, email, or SAP ID with a password.
- FR1.1.2: The system shall allow teachers (supervisors) to log in using username, email, or SAP ID with a password.
- FR1.1.3: The system shall allow administrators to log in using username, email, or SAP ID with a password.
- FR1.1.4: The system shall allow support for role-based access for Student, Teacher (Supervisor), Committee, and Administrator user roles.
- FR1.1.5: The system shall allow creation of authenticated sessions using JWT tokens and storage of active sessions in the database with an expiry time (e.g., 24 hours).
- FR1.1.6: The system shall allow users to log out and invalidate the current session token.
- FR1.1.7: The system shall allow users to request a password reset through email verification using a time-limited reset token.
- FR1.1.8: The system shall allow validation of reset tokens before allowing password reset and prevention of reset token reuse.
- FR1.1.9: The system shall allow authenticated users to change their password by providing the current password.
- FR1.1.10: The system shall allow enforcement of passwords with a minimum of eight characters, including uppercase letters, lowercase letters, numbers, and special characters.
- FR1.1.11: The system shall allow implementation of rate limiting for authentication and password reset endpoints to reduce brute-force attacks.
- FR1.1.12: The system shall allow temporary locking of an account after multiple failed login attempts within a defined time window (e.g., 15 minutes).
- FR1.1.13: The system shall allow maintenance of audit logs for authentication events (login success/failure, logout, account lock, password reset request/completion).
- FR1.1.14: The system shall allow encryption of all authentication traffic using HTTPS in production deployments.

- FR1.2.1: The system shall allow administrators to create user accounts with username, email, SAP ID, role, and an initial password.
- FR1.2.2: The system shall allow administrators to assign a department and maximum supervisees when creating a Teacher (Supervisor) account.
- FR1.2.3: The system shall allow administrators to update user role, active/inactive status, department, maximum supervisees, and reset a user password.
- FR1.2.4: The system shall allow administrators to delete user accounts (with related records handled by database constraints).
- FR1.2.5: The system shall allow administrators to bulk-create users via CSV upload; for Teacher role, the CSV shall include email and department.
- FR1.2.6: The system shall allow administrators to initiate password reset via email for a selected user.
- FR1.2.7: The system shall allow administrators to initiate a security-question challenge and complete password reset after successful verification.
- FR1.2.8: The system shall allow provision of an audit log view for administrators to monitor user actions and security events.

MODULE 2: Profile & Personal Information Management

- FR2.1.1: The system shall allow all authenticated users to view their profile including username, email, SAP ID, role, phone, department, and profile picture.
- FR2.1.2: The system shall allow students to update their name, contact details, and department.
- FR2.1.3: The system shall allow teachers (supervisors) to update their name, contact details, department, research areas, expertise, and availability status.
- FR2.1.4: The system shall allow administrators to update their name, contact details, and department.
- FR2.1.5: The system shall allow validation of profile updates and prevention of duplicate email addresses across users.
- FR2.1.6: The system shall allow all users to upload, update, and delete a profile picture and display it across the system.
- FR2.1.7: The system shall allow maintenance of an audit log of all profile updates and profile picture changes accessible to administrators.
- FR2.1.8: The system shall allow provision of a lookup mechanism to verify user existence by SAP ID or email for workflows that require member validation.

MODULE 3: Project Archive & Search

- FR3.1.1: The system shall allow storage of completed FYP projects with title, abstract, year, department, supervisor, and final grade.
- FR3.1.2: The system shall allow storage of additional optional metadata including technology type, keywords, and student names.
- FR3.1.3: The system shall allow administrators to create, update, and delete archived projects.

- FR3.1.4: The system shall allow administrators to bulk import archived project data using a CSV file.
- FR3.1.5: The system shall allow validation of imported project data for format (title length, abstract length, year range) and completeness, and reporting of per-row failures.
- FR3.2.1: The system shall allow students to search archived projects using keywords, supervisor name, year, department, technology type, or grade.
- FR3.2.2: The system shall allow teachers (supervisors) to search archived projects using the same criteria.
- FR3.2.3: The system shall allow administrators to search archived projects using the same criteria.
- FR3.2.4: The system shall allow support for advanced search with multiple filters using AND/OR operators.
- FR3.2.5: The system shall allow support for pagination and sorting for search results.
- FR3.2.6: The system shall allow display of search results with project summaries and allow users to view full project details.
- FR3.2.7: The system shall allow provision of filter option lists (e.g., departments, years) to support UI search filters.

MODULE 4: Proposal Management

- FR4.1.1: The system shall allow provision of an online proposal submission form for students with draft support (Save as Draft).
- FR4.1.2: The system shall allow the proposal form to include project title, group member details (identified by SAP ID), project description, and proposal PDF upload.
- FR4.1.3: The system shall allow dynamic addition and removal of group members and storage of member information (SAP ID, email, phone, department).
- FR4.1.4: The system shall allow students to upload a proposal PDF for drafts and revision requests and replacement of an existing PDF when re-uploaded.
- FR4.1.5: The system shall allow validation of required fields and PDF upload before allowing proposal submission.
- FR4.1.6: The system shall allow students to select a supervisor from available supervisors before submission and allow display of real-time availability and workload information.
- FR4.1.7: The system shall allow notification of proposal members via email using unique invitation tokens and tracking of acceptance/rejection status per SAP ID.
- FR4.1.8: The system shall allow a proposal to be held in 'pending member confirmation' status until all invited members accept, then submission to the selected supervisor.
- FR4.1.9: The system shall allow a proposal member to accept or reject an invitation using an email link without logging in.
- FR4.1.10: The system shall allow removal of a member from the proposal if they reject the invitation and notification of the lead student.
- FR4.1.11: The system shall allow the lead student to submit a proposal without a rejected member via an external email link.

- FR4.1.12: The system shall allow notification of students via email when supervisors approve, reject, or request revisions, including supervisor feedback when applicable.
- FR4.1.13: The system shall allow hiding of the proposal submission option after supervisor approval.
- FR4.2.1: The system shall allow notification of supervisors via email when a proposal is submitted to them.
- FR4.2.2: The system shall allow supervisors to view proposal details including member list (SAP IDs) and uploaded documents.
- FR4.2.3: The system shall allow supervisors to approve, reject, or request revision of proposals and provision of written feedback.
- FR4.2.4: The system shall allow automatic incrementing of the supervisor workload counter when a proposal is approved.
- FR4.3.1: The system shall allow administrators to upload, update, and remove proposal templates and allow provision of a current template endpoint for the frontend.
- FR4.3.2: The system shall allow logging of all student proposal activities and supervisor proposal actions in proposal activity logs with timestamps.
- FR4.3.3: The system shall allow administrators to view detailed interaction timelines for each proposal.
- FR4.3.4: The system shall allow provision of proposal analytics and reporting including counts by status, by supervisor, recent submissions, and average response time.
- FR4.3.5: The system shall allow administrators to view all proposals with filtering and pagination for administrative monitoring.
- FR4.3.6: The system shall allow restricted access such that administrators cannot approve, reject, or modify student–supervisor proposal decisions through administrative endpoints.

MODULE 5: Supervision Management

- FR5.1.1: The system shall allow supervisors to update their availability status (Available, Busy, Unavailable).
- FR5.1.2: The system shall allow supervisor availability to be reflected in supervisor selection and workload monitoring.
- FR5.2.1: The system shall allow administrators to define and update the maximum number of concurrent projects per supervisor (max supervisees).
- FR5.2.2: The system shall allow tracking of current supervisees and calculation of remaining capacity for each supervisor.
- FR5.2.3: The system shall allow provision of capacity alerts listing supervisors who have reached a defined capacity threshold.
- FR5.2.4: The system shall allow provision of capacity alerts listing supervisors who have exceeded their supervision capacity.
- FR5.2.5: The system shall allow generation of downloadable supervisor workload reports (CSV) for administrative review.
- FR5.2.6: The system shall allow logging of capacity threshold and capacity exceeded events in audit logs for monitoring.

- FR5.2.7: The system shall allow administrators to reset or decrement a supervisor workload counter when required for administrative correction.

MODULE 6: Smart Curriculum & Milestone Configuration (Admin / Coordinator)

- FR6.1.1: The system shall allow support for dynamic academic lifecycle management for FYP-I and FYP-II with week-by-week tracking (Week-1 to Week-16).
- FR6.1.2: The system shall allow coordinators to create department-specific milestone tracks (e.g., Software Engineering, Computer Science, AI, Cyber Security) with automated template assignments.
- FR6.1.3: The system shall allow provision of a 'Smart Milestone Track' based on the student's SAP ID profile, ensuring they only see relevant department tasks.
- FR6.1.4: The system shall allow coordinators to define weekly tasks with 'Intelligent Release Rules' (auto-activating based on the current academic week).
- FR6.1.5: The system shall allow coordinators to upload downloadable templates for weekly tasks with version control support.
- FR6.1.6: The system shall allow coordinators to define instruction-only tasks without templates for conceptual assignments.
- FR6.1.7: The system shall allow provision of 'Smart Deadline Management' with automated countdowns and submission locks.
- FR6.1.8: The system shall allow automatic notification of students and supervisors when a new task is released or a deadline is approaching.
- FR6.1.9: The system shall allow automatic blocking of submissions after deadlines and logging of the event with a 'Late Submission' flag for auditing.
- FR6.1.10: The system shall allow coordinators to reopen deadlines for specific groups with mandatory logged justification and timestamp.
- FR6.1.11: The system shall allow provision of a 'Master Completion Dashboard' for coordinators to monitor group progress across all departments using SAP ID filters.

MODULE 7: Smart Student Weekly Progress & Submission

- FR7.1.1: The system shall allow provision of a 'Personalized Academic Roadmap' for students in read-only mode to visualize future milestones.
- FR7.1.2: The system shall allow enabling of submission access only for the 'Active Window' and the student's assigned track based on SAP ID profile.
- FR7.1.3: The system shall allow display of a 'Smart Deadline Timer' that turns red when the submission window is closing soon.
- FR7.1.4: The system shall allow students to download official templates for physical form tasks with one-click access.
- FR7.1.5: The system shall allow support for 'Draft Mode' for weekly tasks, allowing students to save progress before final submission.
- FR7.1.6: The system shall allow students to upload digital documents for instruction-only tasks with 'Automated File Validation' (format/size check).
- FR7.1.7: The system shall allow binding of each submission to its official template and storage of template reference for audit purposes.

- FR7.1.8: The system shall allow enforcement of strict file format (PDF, DOCX) and maximum file size per task to ensure storage efficiency.
- FR7.1.9: The system shall allow only one active submission per group per task, identifying the group using SAP IDs.
- FR7.1.10: The system shall allow maintenance of 'Submission Version History' (Version Control) for resubmissions allowed by revision requests.
- FR7.1.11: The system shall allow notification of students immediately when a supervisor verifies or requests revision for a submission.

MODULE 8: Smart Supervisor Review & Evaluation

- FR8.1.1: The system shall allow provision of a 'Unified Review Dashboard' for supervisors to track all assigned groups' weekly progress in one view.
- FR8.1.2: The system shall allow supervisors to provide constructive feedback on submissions with 'Quick-Response Templates'.
- FR8.1.3: The system shall allow supervisors to mark submissions as 'Verified' or 'Revision Required' with automated student notification.
- FR8.1.4: The system shall allow storage of detailed evaluation records for each group (linked to SAP IDs), task, and week.
- FR8.1.5: The system shall allow restricted modification of evaluations after final deadlines, except for administrator-authorized corrections with full audit trail.

MODULE 9: Smart Attendance & Performance Digitization

- FR9.1.1: The system shall allow provision of a 'Digital Performance Tracker' for students, linking attendance and progress records to SAP ID.
- FR9.1.2: The system shall allow requirement of 'Supervisor Digital Verification' for attendance submissions, eliminating physical form dependency.
- FR9.1.3: The system shall allow coordinators to mark physical receipt of hardcopy forms if required for regulatory compliance.
- FR9.1.4: The system shall allow locking of attendance submissions after the deadline and storage of the lock event for audit tracking.

MODULE 10: Smart Evaluation & Rubric Enforcement

- FR10.1.1: The system shall allow enforcement of 'Rubric-Based Evaluation' for all graded submissions, ensuring transparency and consistency.
- FR10.1.2: The system shall allow locking of evaluation marks after finalization and preservation of any changes as an 'Audit History' record.
- FR10.1.3: The system shall allow maintenance of complete evaluation history for audit purposes, linked to group SAP IDs, supervisor, and timestamps.

MODULE 11: Smart FYP-I to FYP-II Transition

- FR11.1.1: The system shall allow 'Automatic Carrying Forward' of approved projects from FYP-I into FYP-II with all historical records.

- FR11.1.2: The system shall allow maintenance of continuity by carrying forward supervisors and group members using the same SAP IDs.
- FR11.1.3: The system shall allow prevention of unauthorized project topic changes without coordinator approval, recorded in audit logs.

MODULE 12: Smart Notification & Compliance (Email-Based)

- FR12.1.1: The system shall allow notification via 'Intelligent Notifications' for task releases, upcoming deadlines, and submission outcomes.
- FR12.1.2: The system shall allow notification of students and supervisors regarding approvals, rejections, and revision requests in real-time.
- FR12.1.3: The system shall allow notification of coordinators regarding 'Compliance Gaps' (missing or overdue submissions) with SAP ID search support.

MODULE 13: Smart Security & Role Governance

- FR13.1.1: The system shall allow enforcement of strict 'Role-Based Access Control (RBAC)' across all modules.
- FR13.1.2: The system shall allow implementation of 'Security Violation Detection', recording unauthorized access attempts for monitoring.
- FR13.1.3: The system shall allow sensitive actions (password resets, role changes, workload edits) to be recorded in 'High-Integrity Audit Logs' with user SAP ID.

MODULE 14: Smart Audit, Archive & Knowledge Base

- FR14.1.1: The system shall allow archiving of completed projects with 'Intelligent Metadata' (reports, evaluations, student SAP IDs).
- FR14.1.2: The system shall allow support for 'Semantic Search' in the archive (title, year, department, supervisor, technology keywords) to support knowledge reuse.
- FR14.1.3: The system shall allow archived projects to be accessible through the 'Project Archive Module' for authorized academic roles.